Outlier Detection & Visual Diagnostics

Workshop 4 — Module 5

Prof.Asoc. Endri Raco

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Data Visualization for Data Scientists
UNYT Tirana

Module Roadmap

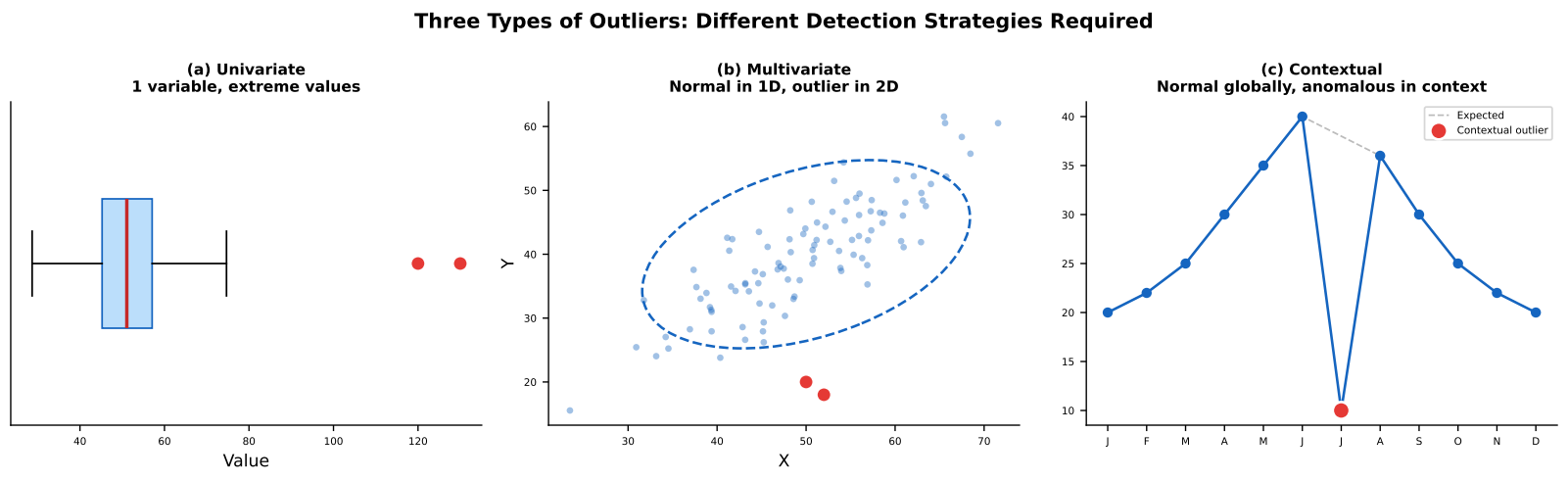
1. Outlier taxonomy: univariate, multivariate, contextual
2. IQR fences and the boxplot as a detection tool
3. Z-score vs IQR: parametric vs robust detection
4. Mahalanobis distance: multivariate outlier detection
5. Regression diagnostics: leverage, influence, Cook's D
6. What to do with outliers: cause determines action

Learning Outcome

You will detect outliers using both robust (IQR) and parametric (z-score, Mahalanobis) methods, diagnose influential observations in regression with Cook's distance, and make principled decisions about whether to keep, flag, or remove each outlier.

Outlier Taxonomy

Three Types of Outliers

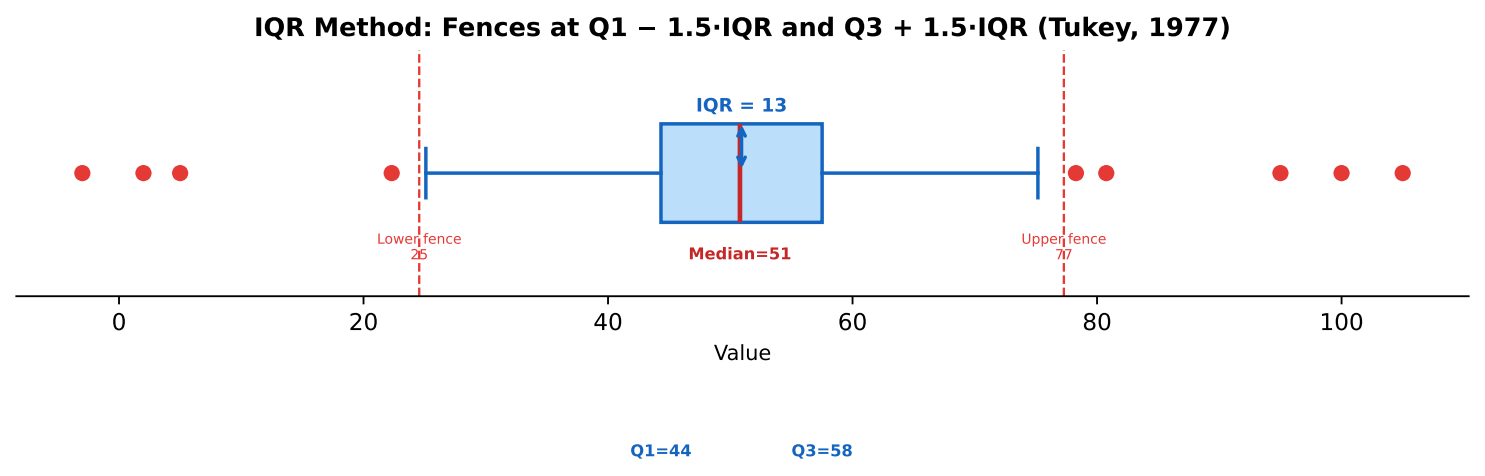


Pro-Tip

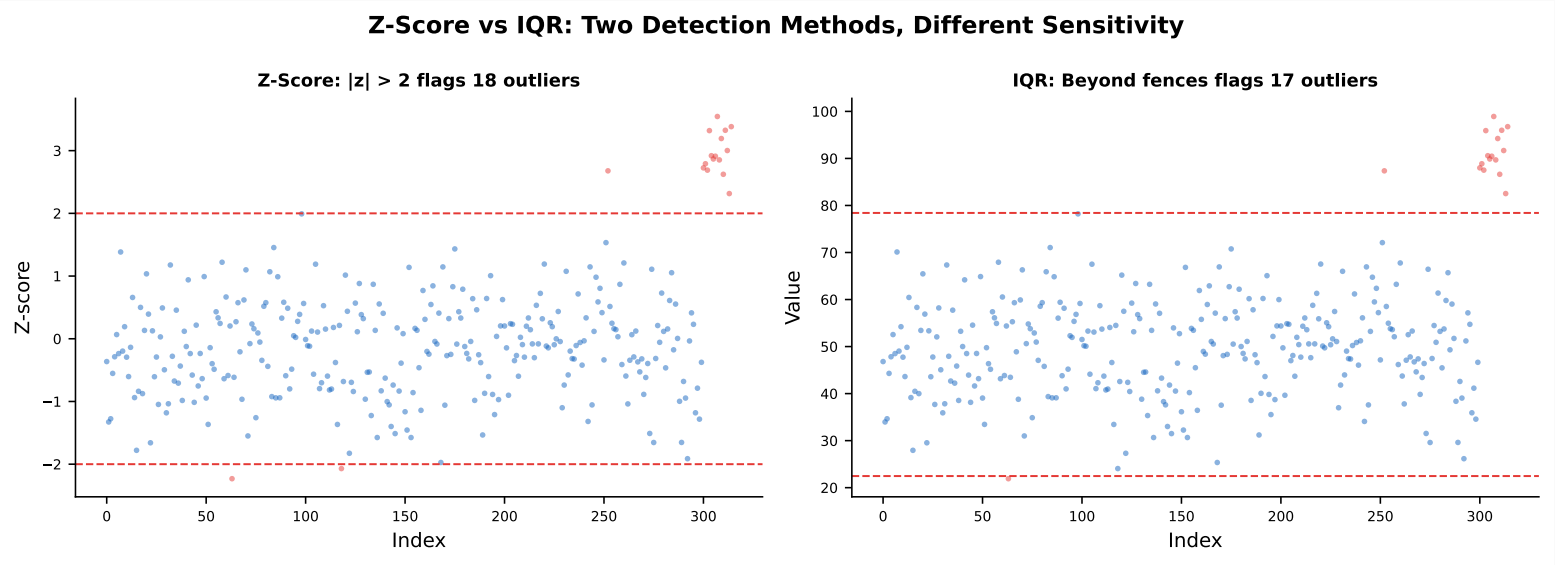
A point normal in each variable individually can still be a **multivariate outlier** (panel b). And a point normal globally can be a **contextual outlier** — anomalous only in its local context (panel c). Different types require different detection strategies.

IQR & Z-Score Detection

The IQR Method: Tukey's Fences



Z-Score vs IQR



Data Insight

Z-score assumes normality: $z = (x - \bar{x})/s$. If the data is skewed, the mean and SD are pulled by outliers, making z-scores unreliable. IQR is **robust**: it uses medians and quartiles, which are resistant to extreme values. Default to IQR for EDA; use z-scores only for approximately normal data.

Detection in Code

R

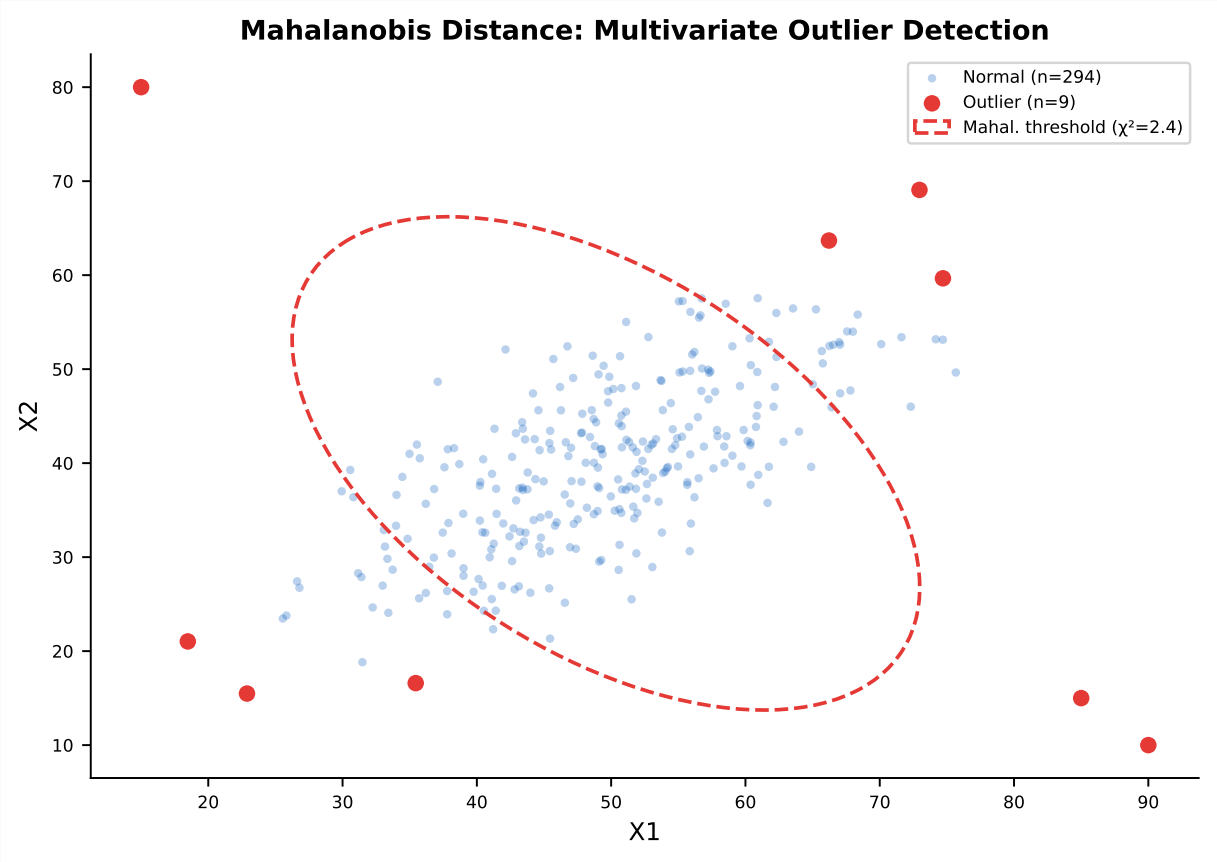
```r
# IQR method
q1 <- quantile(x, 0.25)
q3 <- quantile(x, 0.75)
iqr <- q3 - q1
outlier <- x < q1 - 1.5*iqr |
           x > q3 + 1.5*iqr

# Built into boxplot
ggplot(df, aes(y = Rating)) +
  geom_boxplot(outlier.colour =
    "#E53935", outlier.size = 2)

# Z-score
z <- scale(x)  # standardise
outlier_z <- abs(z) > 2
```

Python

```python
# IQR method
q1 = np.percentile(x, 25)
q3 = np.percentile(x, 75)
iqr = q3 - q1
outlier = (x < q1 - 1.5*iqr) | \
          (x > q3 + 1.5*iqr)

# Built into boxplot
ax.boxplot(rating,
    flierprops=dict(
        markerfacecolor="#E53935"
            ))

# Z-score
from scipy.stats import zscore
z = zscore(x)
outlier_z = np.abs(z) > 2
```

Multivariate Detection

Mahalanobis Distance



Data Insight

Mahalanobis distance measures how far a point is from the centre of the data cloud, accounting for correlations. A point can be within normal range on each axis separately but far from the elliptical cloud. Threshold: $\chi^2(p, 0.95)$ where p = number of variables.

Multivariate Detection in Code

R

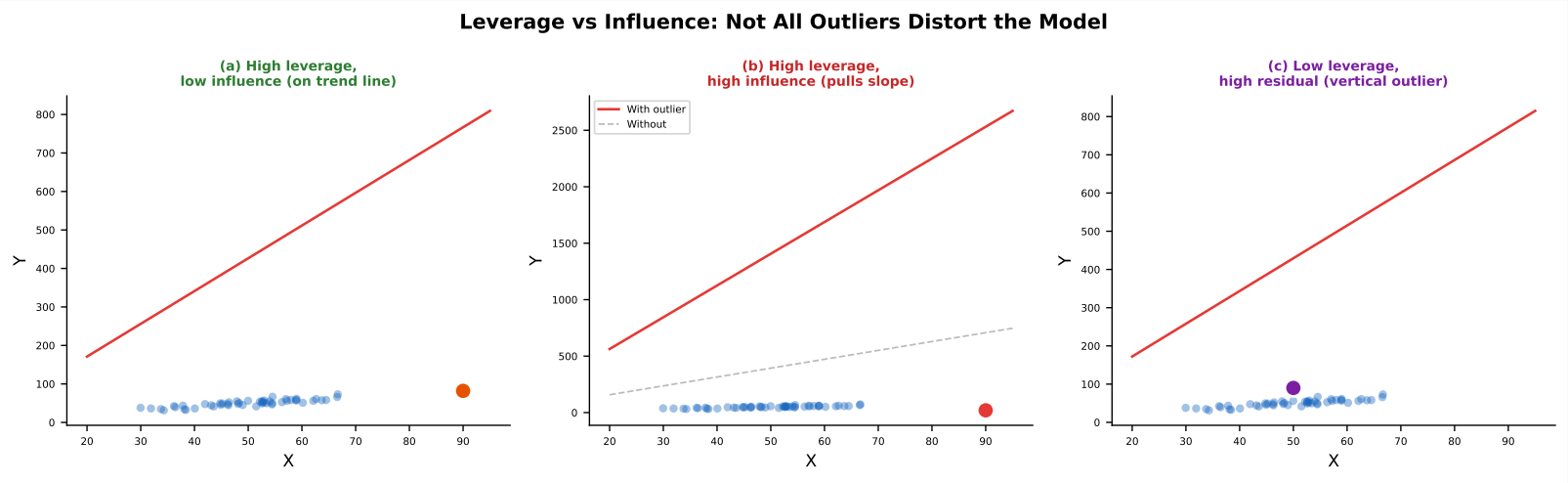
```r
# Mahalanobis (base R)
md <- mahalanobis(
  df[, c("x1", "x2")],
  colMeans(df[, c("x1", "x2")]),
  cov(df[, c("x1", "x2")]))
threshold <- qchisq(0.95, df = 2)
outlier <- md > threshold

# Robust (MCD estimator)
library(robustbase)
mcd <- covMcd(df[, c("x1","x2")])
md_robust <- mahalanobis(
  df[,c("x1","x2")],
  mcd$center, mcd$cov)
```

Python

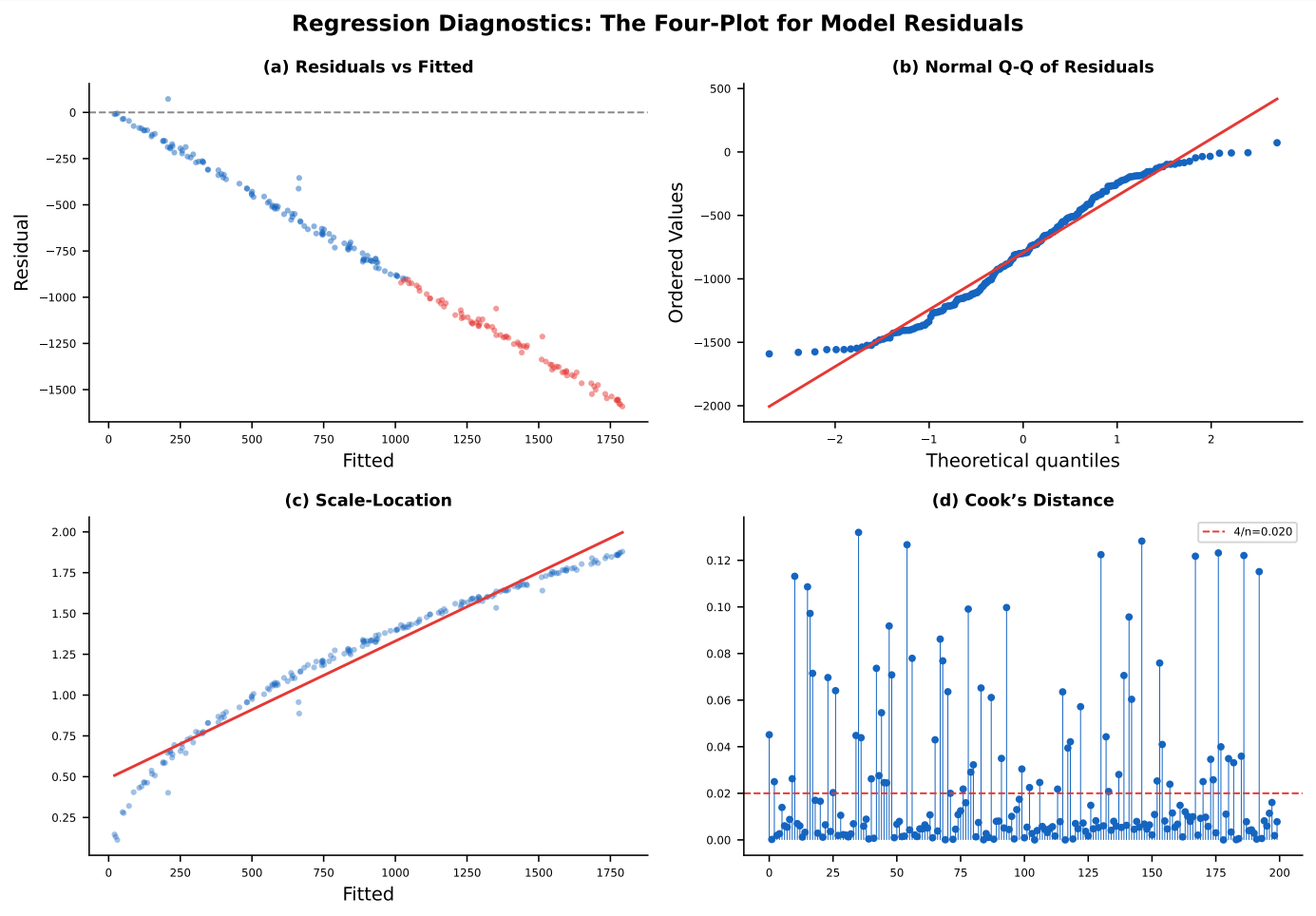
```python
from scipy.spatial.distance
     import (
    mahalanobis)
mu = np.mean(X, axis=0)
cov = np.cov(X.T)
cov_inv = np.linalg.inv(cov)
md = [mahalanobis(xi, mu, cov_inv
     )
      for xi in X]
threshold = np.sqrt(
    chi2.ppf(0.95, df=2))

# Robust: EllipticEnvelope
from sklearn.covariance import (
    EllipticEnvelope)
clf = EllipticEnvelope(
    contamination=0.05)
labels = clf.fit_predict(X)
# -1 = outlier, 1 = inlier
```

Regression Diagnostics

Leverage vs Influence



The Four Diagnostic Plots for Regression



What Each Diagnostic Reveals

Plot	What It Checks	Bad Sign
(a) Resid. vs Fitted	Linearity, homoscedasticity	Funnel, curve
(b) Q-Q of Residuals	Normality of errors	S-shape, heavy
(c) Scale-Location	Constant variance (spread vs level)	Upward trend
(d) Cook's Distance	Influence on fitted model	Spikes $> 4/n$

Pro-Tip

In R: `plot(lm_model)` produces all four automatically. In Python: `statsmodels.graphics.plot_regress_exog()` or manual construction. Cook's $D > 4/n$ flags influential points. High leverage + high residual = high influence.

Regression Diagnostics in Code

R

```r
model <- lm(Rating ~ log(Reviews)
    ,
  data = apps_clean)

# All 4 plots at once
par(mfrow = c(2, 2))
plot(model)

# Or with broom + ggplot
library(broom)
diag <- augment(model)
# diag has: .fitted, .resid,
#   .hat (leverage), .cooksd

ggplot(diag, aes(x = .fitted,
    y = .resid)) +
  geom_point(alpha = 0.3) +
  geom_hline(yintercept = 0,
    linetype = "dashed") +
  geom_smooth(se = FALSE,
    color = "#E53935")
```

Python

```python
import statsmodels.api as sm

X = sm.add_constant(
    np.log(apps["Reviews"]+1))
model = sm.OLS(apps["Rating"],
    X).fit()

# Influence measures
infl = model.get_influence()
cooks = infl.cooks_distance[0]
leverage = infl.hat_matrix_diag

# Manual diagnostic plots
fig, axes = plt.subplots(2, 2)
axes[0,0].scatter(
    model.fittedvalues,
    model.resid, s=5, alpha=0.3)
# ... Q-Q, Scale-Loc, Cook's ...

# Quick: plot_regress_exog
from statsmodels.graphics import
    (
    regressionplots as rp)
rp.plot_regress_exog(model, 1)
```

What to Do

Cause Determines Action

What to Do with Outliers: Cause Determines Action

Cause	Action	Example
Data entry error	Correct or remove	Impossible values (Rating = 19, Age = -5)
Measurement error	Flag, sensitivity analysis	Sensor glitch, rounding artefact
Rare but real	Keep, report separately	Extreme genuine obs (billionaire income)
Different population	Segment, analyse apart	Mixture of groups (Paid apps in Free-dominated)
Influential point	Fit with and without; report both	Changes regression slope significantly

The "Fit With and Without" Protocol

For any influential outlier, run the analysis **twice**:

Step	Action
1	Fit model with all data
2	Identify influential points (Cook's $D > 4/n$)
3	Refit model **without** those points
4	Compare coefficients and R^2
5	Report both results

If conclusions change $\Rightarrow$ the finding depends on a few points. This is a **fragility signal**: the result is not robust. Report this honestly rather than silently deleting outliers.

Synthesis

Key Takeaways

1. Three outlier types: **univariate** (extreme on one axis), **multivariate** (normal per-variable, extreme jointly), **contextual** (anomalous in local context only).
2. **IQR fences** (Tukey) are robust to skew; **z-scores** assume normality. Default to IQR for EDA.
3. **Mahalanobis distance** detects multivariate outliers by measuring distance from the data cloud's elliptical centre.
4. **Leverage** $\neq$ **influence**. High leverage + high residual = high influence (Cook's D). Check with `plot(lm())` in R or `get_influence()` in Python.
5. **Never delete outliers blindly**. Determine the cause first: data error $\rightarrow$ fix; real extreme $\rightarrow$ keep; influential $\rightarrow$ fit with and without.

Essential Readings

- Tukey, J. W. (1977). *Exploratory Data Analysis*, Chapter 2 (Box-and-Whisker Plots, Fences).
- Rousseeuw, P. J. & van Zomeren, B. C. (1990). "Unmasking multivariate outliers." *JASA*, 85(411).
- Cook, R. D. (1977). "Detection of influential observations in linear regression." *Technometrics*, 19(1).
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapter 9 (Robust Statistics).

Next Module: Correlation & Association Visualization (W04-M06)